STAT2005 Computer Simulation

Lecture 5 — Simulation of Continuous Random Variables

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26 Mar 2026

Goals today

- **Part 1:** Transformation Methods (Extended)
  - Transformation of a Single Random Variable
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    - Continuous Case
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    - Exponential Distribution Example (Closed-form Inverse CDF)
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  - Normal-Cauchy Example
- **Part 3:** Precision of Simulation Result (Revision)
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Part 1: Transformation of Random Variables

Recap: Where We Are So Far

- We use **simulation** to study random systems by repeating experiments many times
- To simulate, we need to generate **random variables**
- Random variables follow **probability distributions**

The Key Building Block

- Computers generate **pseudo-random numbers**
- In practice, we generate reproducible rvs: $\boxed{U \sim \text{Uniform}(0,1)}$
- Then, we used **simple rules** to generate random variables:

Bernoulli(p)

Generate $U \rightarrow$ If $U < p$, return 1, else return 0

Discrete Distributions:

Use the precision method or the inverse CDF method

- **Idea:** Start with Uniform(0,1), then **transform it.**
- How do we generate **any other distribution**? Exponention? Normal? Custom distributions?

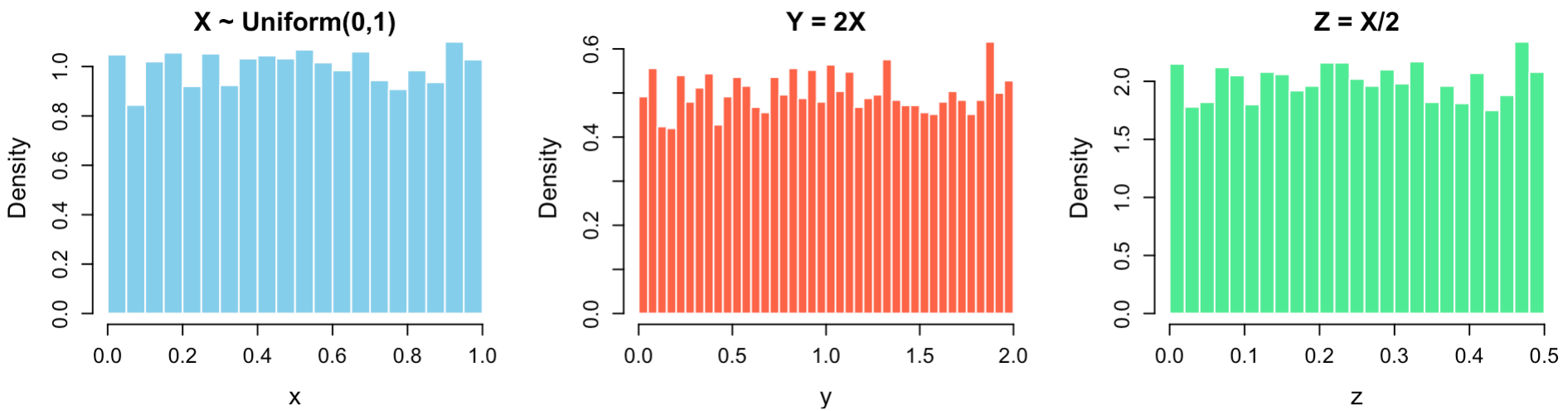
Transformation of a Single Random Variable

When we apply a function to a random variable, the result is itself a new random variable. If X has a known distribution and we define

$$Y = g(X),$$

then the key question is: **What is the distribution of Y ?**

- Suppose $X \sim \mathrm{Uniform}(0,1)$ and $Y = 2X, Z = X/2.$



- If $Y = 2X \rightarrow$ values spread out (**stretched**) $\rightarrow$ density becomes lower.
- If $Y = X/2 \rightarrow$ values squeezed (**compressed**) $\rightarrow$ density becomes higher

🎯 **Key idea:** Probability is **conserved**, but density is **redistributed**

Discrete Case (Intuition)

When we transform a random variable, $Y = g(X)$, multiple values of X may map to the same value of Y.

The probability of Y is obtained by **collecting contributions** from all such values:

$$p_Y(y) = \sum_{x:g(x)=y} p_X(x).$$

This is a "*probability bookkeeping*" rule: every value of Y inherits probability from the X-values that produce it.

Example: Let $X \in \{1, 2, 3, 4\}$ with given probabilities

$$P(X = 1) = 0.1, \quad P(X = 2) = 0.2,$$
$$P(X = 3) = 0.3, \quad P(X = 4) = 0.4.$$

Define:

$$Y = \begin{cases} 0 & \text{if } X \text{ is odd} \\ 1 & \text{if } X \text{ is even} \end{cases}$$

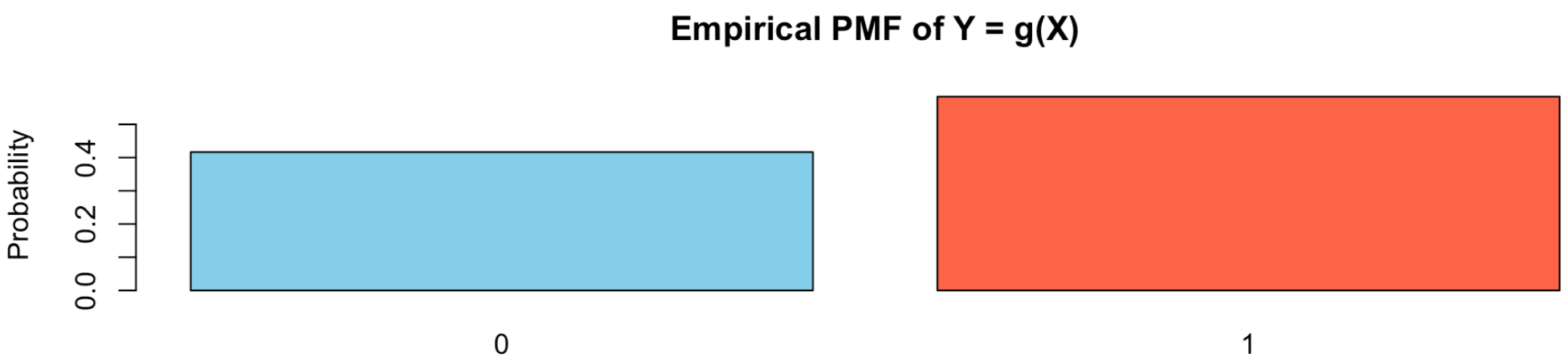
Then:

$$Y = 0 : \text{from } X = 1, 3 \qquad Y = 1 : \text{from } X = 2, 4$$

PMF of Y:

$$P(Y = 0) = P(X = 1) + P(X = 3) = 0.1 + 0.3 = 0.4,$$
$$P(Y = 1) = P(X = 2) + P(X = 4) = 0.2 + 0.4 = 0.6$$

```r
set.seed(1234)

# Original discrete variable
X <- sample(1:4, size = 5000, replace = TRUE,
            prob = c(0.1, 0.2, 0.3, 0.4))

# Transformation
Y <- ifelse(X %% 2 == 1, 0, 1)

# Plot empirical pmf of Y
barplot(prop.table(table(Y)),
        col = c("skyblue", "tomato"),
        main = "Empirical PMF of Y = g(X)",
        ylab = "Probability")
```



Continuous Case

- For **discrete** variables: we **add probabilities.**
- For **continuous** variables: $\boxed{P(X = x) = 0.}$

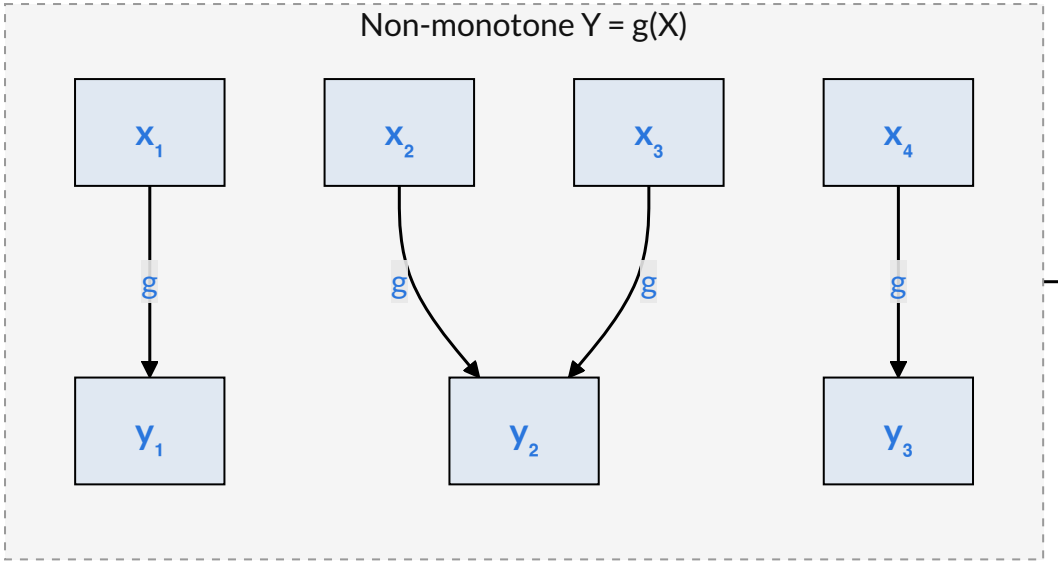
👉 We cannot "collect probabilities" anymore.

- For continuous random variables, there are in principle two approaches:

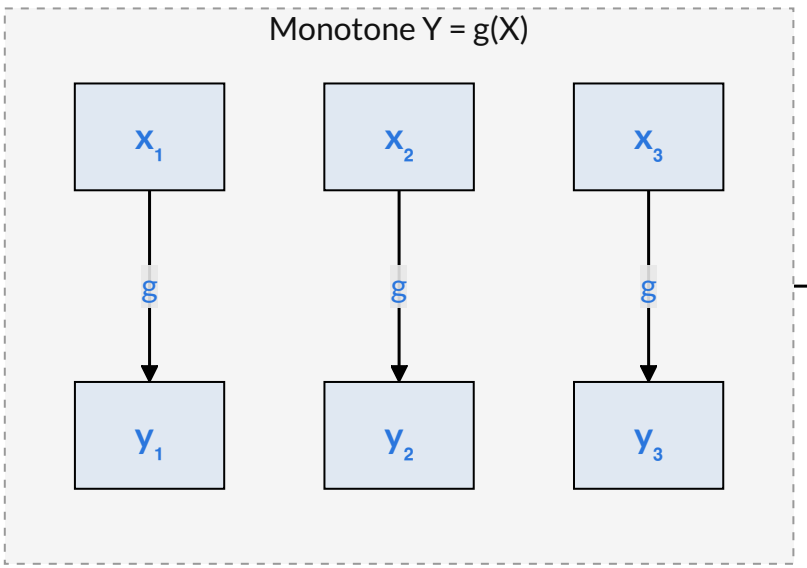
CDF-based Method	PDF-based Method
derives the distribution of Y by working with cumulative probabilities	determines how density changes under the transformation

Key Idea: Density Transformation

- Instead of probabilities, we work with **density.**
- When we transform using $Y = g(X)$, we must track **how density is stretched or compressed.**
  - stretches some regions → density decreases
  - compresses some regions → density increases
- The adjustment is captured by a **derivative.**



A y may have multiple preimages



Each y has exactly one preimage

Case 1: Monotone Transformations

- If g is **one-to-one** and **strictly increasing or decreasing**, each value of Y comes from exactly one X.
- Then, the PDF of Y is given by the change-of-variables formula,

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

- The derivative term adjusts for **stretching/compression.**

Example: Let $X \sim \mathcal{N}(0,1)$ and define $Y = 3X + 2$. Determine the PDF of Y.

We know $Y = g(X) = 3X + 2$, which is strictly increasing. The inverse is

$$x = g^{-1}(y) = \frac{y-2}{3} \quad \text{and} \quad \frac{d}{dy} g^{-1}(y) = \frac{1}{3}.$$

The PDF of X is

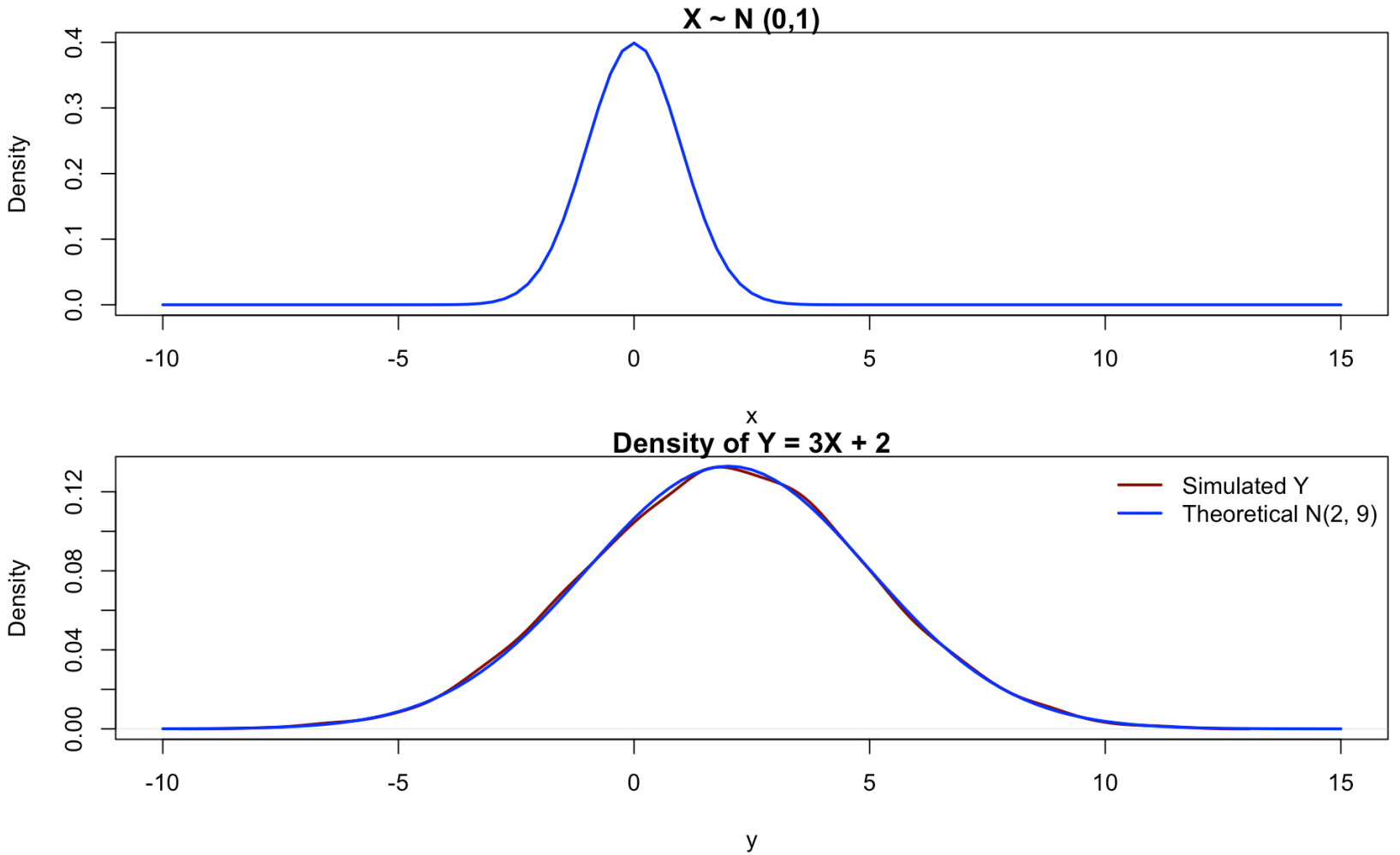
$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}.$$

By the change-of-variables formula,

$$f_Y(y) = f_X\left(\frac{y-2}{3}\right) \cdot \frac{1}{3} = \frac{1}{3} \cdot \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{y-2}{3}\right)^2\right).$$

This is recognised as the PDF of $\boxed{Y \sim \mathcal{N}(2,9)}$.

- Suppose $X \sim N(0,1)$ and $Y = 3X + 2$



- Because $Y = 3X + 2$ stretches the values of X, the density becomes more spread out (lower peak).
- To preserve total probability, the density is scaled by the derivative term in the change-of-variables formula:

$$\left| \frac{dx}{dy} \right| = \frac{1}{3}.$$

Aside: Change-of-variables Formula (Intuition)

Key Idea: Probability is preserved under transformation, i.e.

$$P(a \leq Y \leq b) = P(g^{-1}(a) \leq X \leq g^{-1}(b))$$

To understand how density changes, we look at a **small interval**:

$$P(y \leq Y \leq y + dy) = P(x \leq X \leq x + dx), \tag{1}$$

where $\boxed{x = g^{-1}(y)}$. From the local linear approximation (calculus), we have

$$dx \approx \frac{dx}{dy} dy.$$

For very small intervals, probability is approximately: **(probability ≈ density × width)**. Then,

$$\begin{aligned} P(y \leq Y \leq y + dy) &\approx f_Y(y)\, dy, \\ P(x \leq X \leq x + dx) &\approx f_X(x)\, dx. \end{aligned} \tag{2}$$

From (1) and (2), we have

$$\begin{aligned} f_Y(y)\, dy &= f_X(x)\, dx \\ f_Y(y) &= f_X(x) \frac{dx}{dy}, \end{aligned}$$

and since densities must be positive: $\boxed{f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|}$

Takeaway: The derivative term tells us how much the transformation stretches or compresses the space.

Case 2: Non-Monotone Transformations

- If g is *not* one-to-one, multiple values of X map to the same Y.
- Then we must **add all contributions**:

$$f_Y(y) = \sum_{x:g(x)=y} f_X(x) \left| \frac{dx}{dy} \right|$$

Example: Let $X \sim \mathcal{N}(0,1)$ and define $Y = X^2$. Find the PDF of Y.

- For a given y, two values map to it:

$$x = \sqrt{y}, \quad x = -\sqrt{y}$$

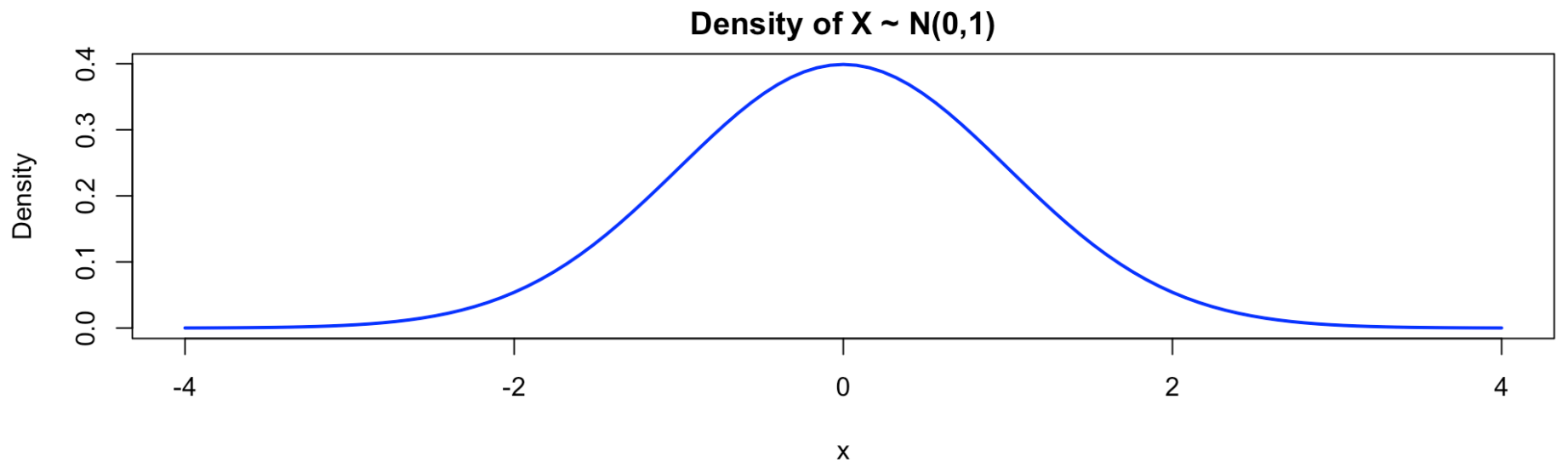
- Both contribute to the density
- Using the non-monotone trasformation rule, we have:

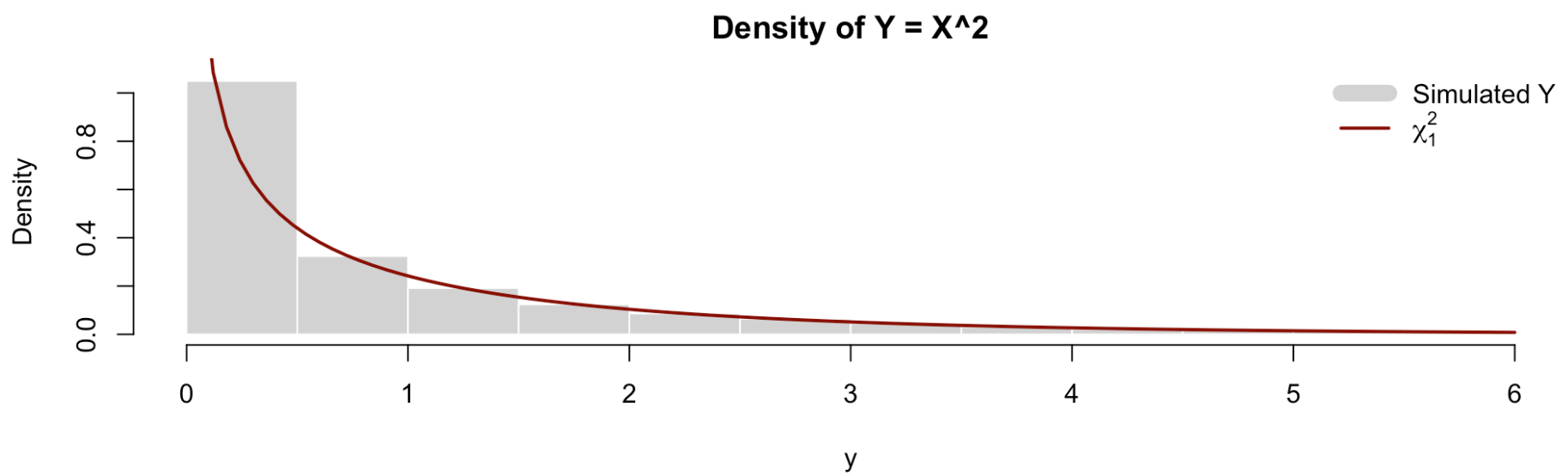
$$f_Y(y) = \frac{1}{\sqrt{2\pi}} e^{-y/2} \cdot \frac{1}{\sqrt{y}}.$$

- This is recognised as the PDF of a Chi-squared distribution with 1 degree of freedom, i.e. $\boxed{Y \sim \chi_1^2}$

See **Transformation of Random Variables** course note for a full derivation of $f_Y(y)$.

▶ Suppose X ~ N(0,1) and Y = X^2





- $X \sim N(0,1)$ distribution spreads across both negative and positive values.
- $Y = X^2$ distribution is right-skewed (non-negative).

Why Transformation Matters to Simulation

So far:

We start with X → Apply transformation $Y = g(X)$ → Find distribution of Y

Now we reverse the idea:

Start with $U \sim \text{Uniform}(0, 1)$ → Choose g to create a desired distribution

👉 **Goal:**

$$X = g(U)$$

so that X has the distribution we want

How do we find such a transformation g ?

Use the **CDF**

This leads to

Inverse Transform Sampling

Inverse Transform Sampling

Every random variable has a CDF:

$$F_X(x) = P(X \leq x)$$

The CDF maps values of X → probabilities in (0,1)

> 👉 This suggests a connection to **Uniform(0,1)**

Probability Integral Transform

If X has CDF F_X, then:

$$U = F_X(X) \sim \text{Uniform}(0,1)$$

> 👉 The CDF transforms **any distribution** → **Uniform(0,1)**

then reversing gives:

$$\boxed{X = F_X^{-1}(U), \quad U \sim \text{Uniform}(0,1)}$$

which is known as the **sampling rule**.

To simulate $X \sim F_X$:

Generate $U \sim \text{Uniform}(0,1)$ → **Transform:** $X = F_X^{-1}(U)$

Exponential Distribution Example (Closed-form Inverse CDF)

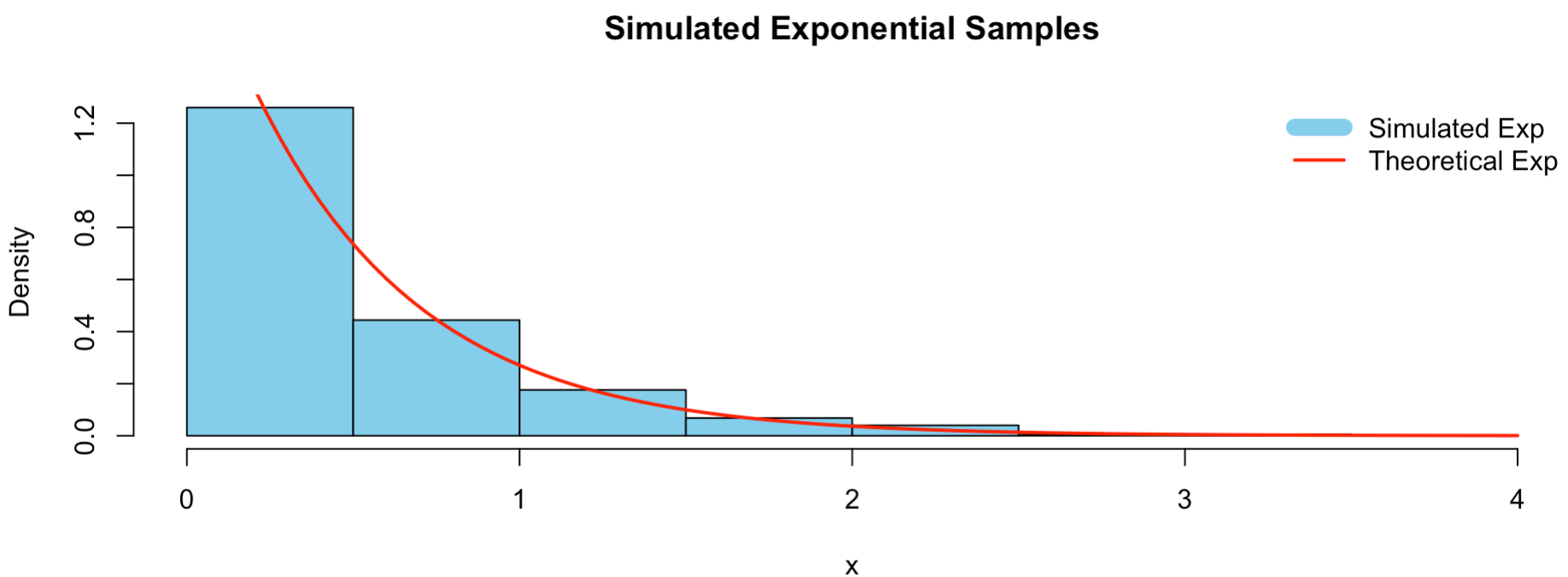
Let $X \sim \mathrm{Exp}(\lambda)$ with the following CDF:

$$F(x) = 1 - e^{-\lambda x}$$

Setting $U = F(X)$ and solving for X gives the inverse term:

$$X = -\frac{1}{\lambda}\log(1-U), \quad U \sim \mathrm{Uniform}(0,1).$$

- X ~ Exp($\lambda = 2$)



Quadratic CDF Example

> Consider the **electrical current distribution** model where the pdf of X is given by
> $$f(x) = 1.25 - 0.25x, \quad 2 \le x \le 4.$$
> Suppose a simulation of X is required as part of some larger system analysis.

Step 0: Check for valid PDF

$$\int_2^4 (1.25 - 0.25x)\, dx = 1 \quad \rightarrow \quad \text{valid density.}$$

Step 1: Determine the CDF $F(x)$

$$\begin{aligned} F(x) &= \int_2^x (1.25 - 0.25y)\, dy \\ &= \left[1.25y - 0.125y^2\right]_2^x \\ F(x) &= 1.25x - 0.125x^2 - 2 \end{aligned}$$

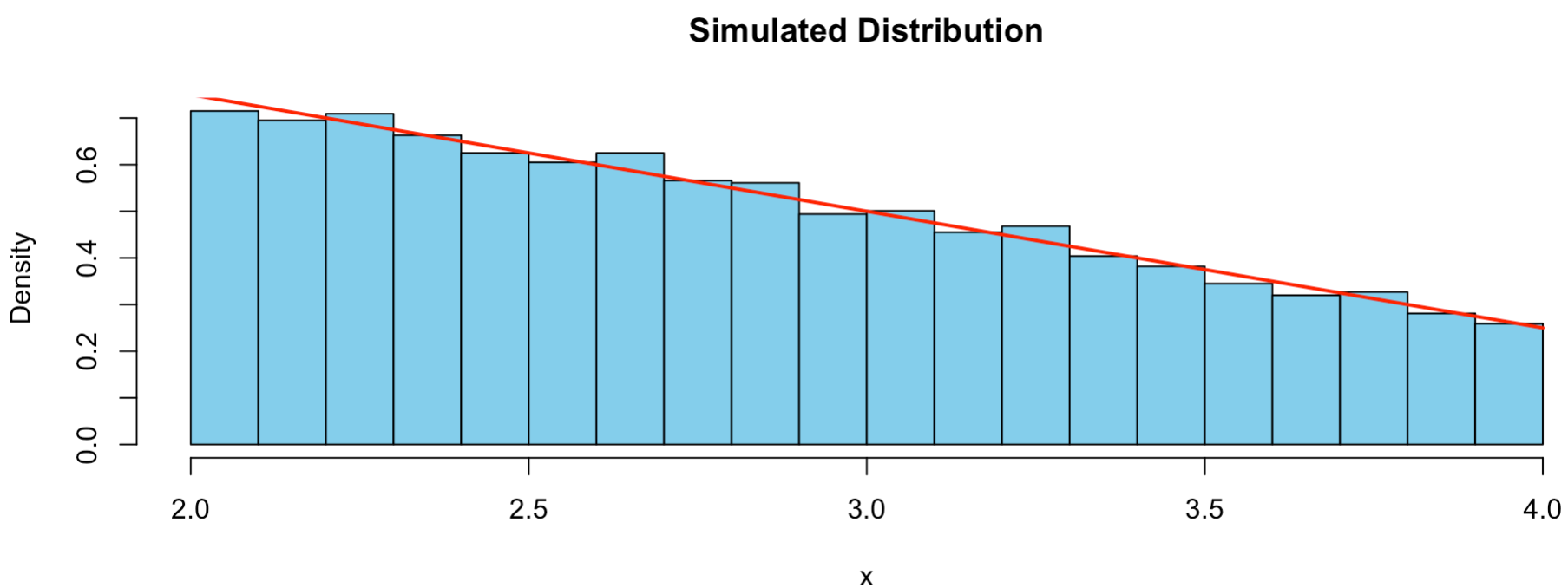
Step 2: Determine the inverse CDF $X = F_X^{-1}(U)$

$$\begin{aligned} u &= F(x) = 1.25x - 0.125x^2 - 2 \\ 0 &= x^2 - 10x + 16 + 8u \\ x &= 5 \pm \sqrt{9 - 8u} \end{aligned}$$

- Since the support is $2 \le x \le 4,$ the only root satisfies this is $\boxed{x = 5 - \sqrt{9 - 8u}}$.

- $\boxed{X = 5 - \sqrt{9 - 8U}}$.
- This transformation maps Uniform(0,1) values into the interval [2,4] with the correct shape.

```r
set.seed(1234)
u <- runif(10000)
x <- 5 - sqrt(9 - 8 * u)

hist(x, prob = TRUE, col = "skyblue",
     main = "Simulated Distribution")

curve(1.25 - 0.25*x, add = TRUE, col = "red", lwd = 2)
```



Normal Distribution Example (No Closed-form Inverse CDF)

Recall the PDF of the **standard normal distribution** $X \sim N(0,1)$:

$$f(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right), \quad x \in \mathbb{R}.$$

The CDF of X is $\Phi(x)$:

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} \exp\left(-\frac{y^2}{2}\right) dy,$$

has <u>no closed-form expression</u>.

- The CDF has no closed-form expression, so neither does its inverse.
  - We must compute it numerically.
- This is the reason why we use software or table to calculate Normal probabilities.

Cubic Polynomial CDF Example

$F(x)$ is a **polynomial function.**

The measurement error X (in mV) of a particular volt-meter has the following distribution:

$$f(x) = \begin{cases} \dfrac{4 - x^2}{9}, & -1 \leq x \leq 2 \\ 0, & \text{otherwise.} \end{cases}$$

The CDF is

$$F(x) = \frac{-x^3 + 12x + 11}{27}, \quad -1 \leq x \leq 2,$$

which is a cubic polynomial function.

- It is not easy to solve for $x = F^{-1}(u)$.
- Numerical methods (e.d. Newton–Raphson) should be used to solve this kind of problem

When Does Inverse Transform Work?

Even if we *can* solve for inverse CDF, it may be:

- slow
- unstable
- impractical in simulation

Case	Difficulty
Exponential	✔ Easy (closed-form inverse)
Electrical example	✔ Moderate (quadratic)
Normal	❌ No closed-form inverse
Polynomial CDF	⚠️ Requires numerical methods

Note

👉 Not all distributions are equally tractable.

Limitations of Inverse Transform

No closed-form inverse

Many distributions (e.g. Normal) do not have an explicit inverse CDF.

Inversion can be difficult

Even when $F(x)$ is known, solving $u = F(x)$ may require numerical methods.

Computational cost

Numerical inversion can be slow or impractical in large simulations.

👉 **We need more general methods** that does not reqire F^{-1} and work for complex distributions.

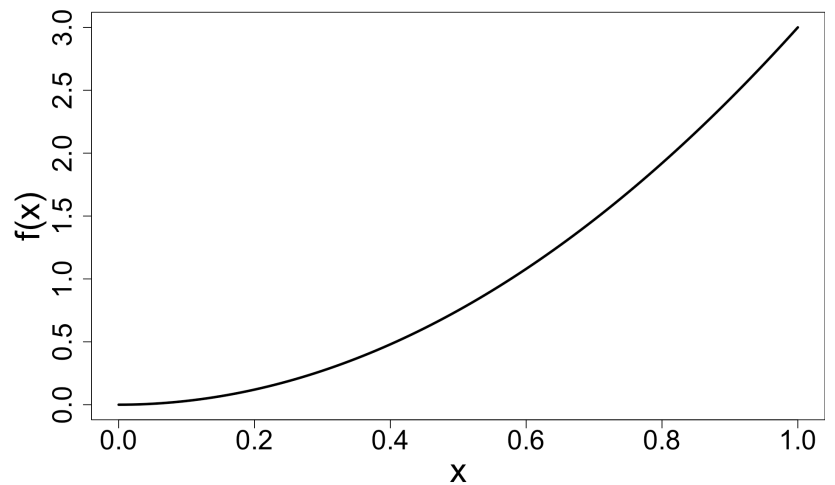
- **Acceptance–Rejection Sampling**
- Specialised transformations such as **Box–Muller** for Normal simulation

Part 2: The Acceptance-Rejection Methods

Acceptance-Rejection Sampling

> **Example:** We want to simulate
> $$f(x) = 3x^2, \quad 0 \le x \le 1,$$
> which has increasing density with more weight near 1.
>
> How do we simulate this?

```r
x <- seq(0, 1, length.out=100)
par(mar=c(4,5,1,0))
curve(3 * x^2, type = "l",
      lwd = 3, ylab = "f(x)",
      cex.axis=2, cex.lab=2.5)
```



First, check for valid PDF.

$$\int_0^1 3x^2\,dx = 1 \rightarrow \text{ valid PDF.}$$

Why Not Inverse Transform?

- We would need $X = F^{-1}(U)$.
- Finding $F(x)$ is easy, but inverting it is not simple.
- 👉 Let's try a different idea.

Key Idea

- Instead of transforming, we **sample points in a region.**

Draw a random point (X, Y) → Accept it if it lies under the curve

- 👉 This turns sampling into a **geometry problem.**

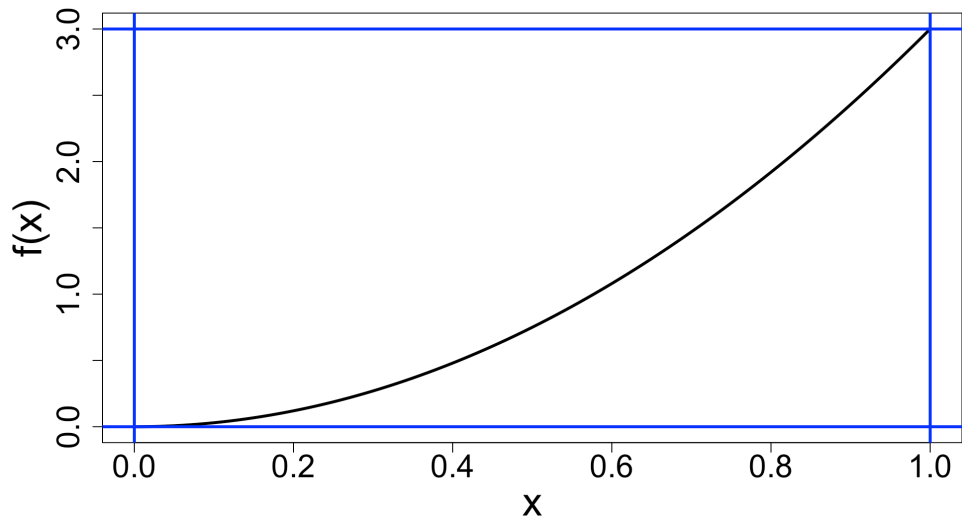
Visualisation

Consider the rectangle:

$$0 \le x \le 1, \quad 0 \le y \le 3.$$

The curve $y = 3x^2$ sits inside this box.

👉 Accept points under the curve
👉 Reject points above it



Acceptance–Rejection Algorithm

1. Generate:
   - $X \sim \text{Uniform}(0, 1)$
   - $Y \sim \text{Uniform}(0, 3)$
2. If:

$$Y \le 3X^2$$

→ Accept X

3. Otherwise → Reject and repeat

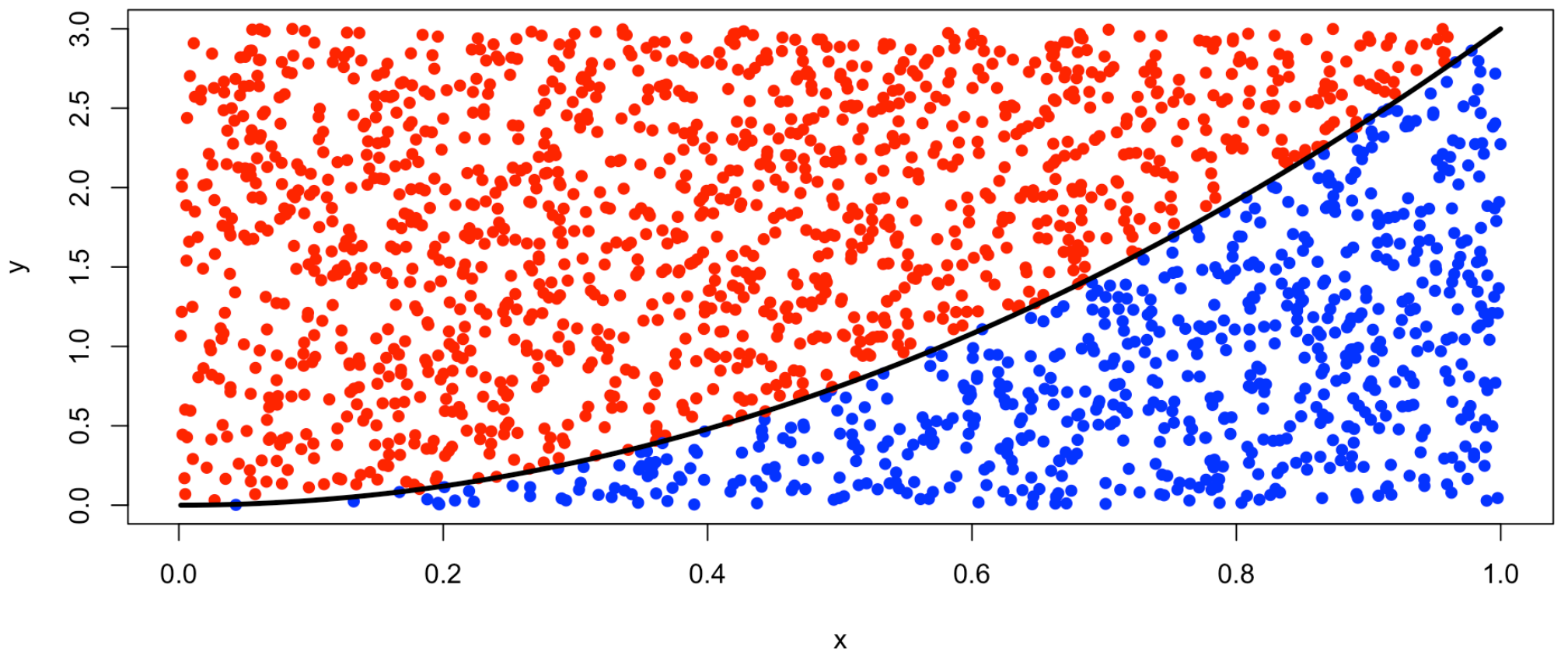
Why Does This Work?

- Points are uniformly distributed in the rectangle.
- Probability of acceptance:

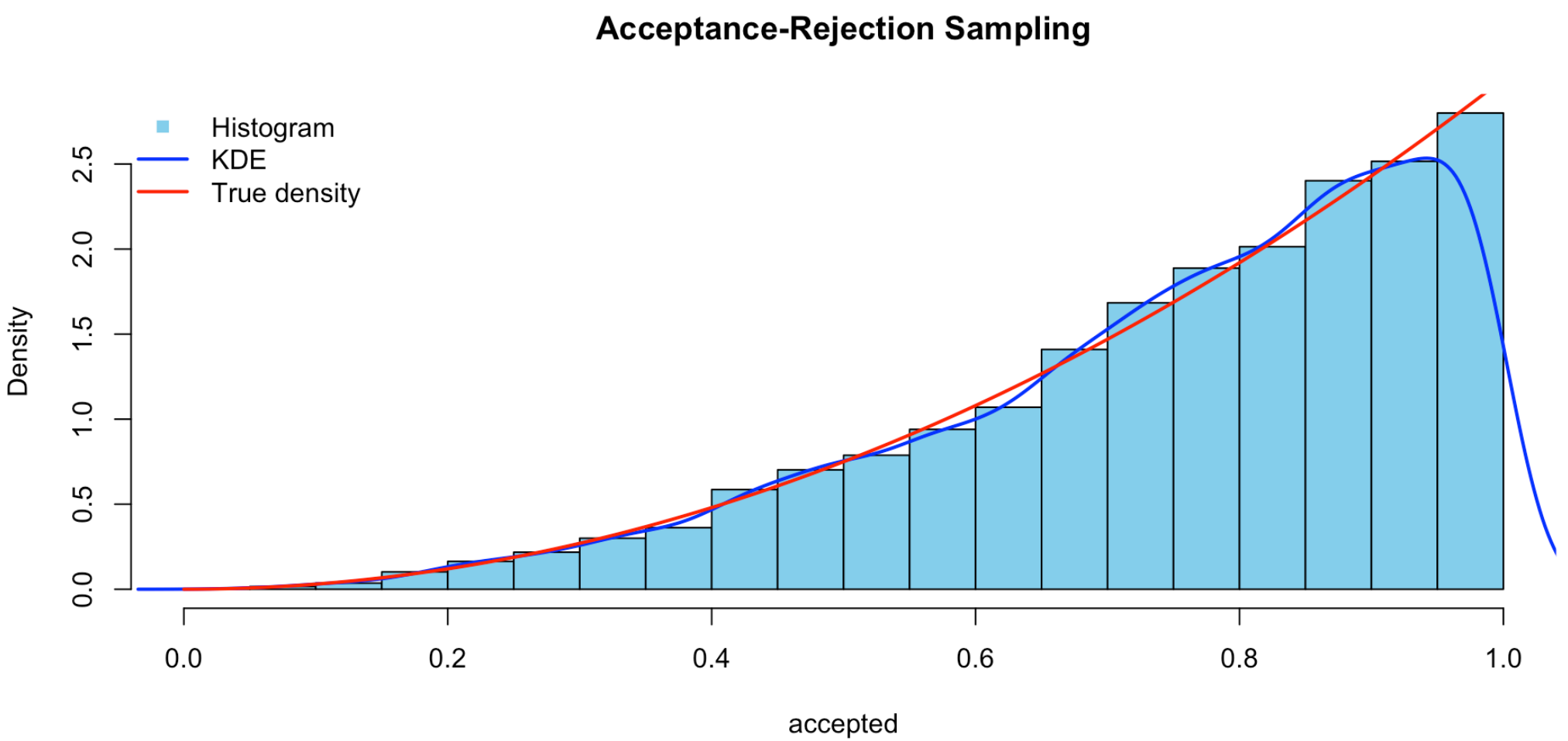
> proportional to area under the curve.

👉 Accepted X values follow the target distribution.

```r
# Visual points
x <- runif(2000)
y <- runif(2000, 0, 3)

plot(x, y, pch = 16, col = ifelse(y <= 3*x^2, "blue", "red"))
curve(3*x^2, add = TRUE, col = "black", lwd=3)
```



```r
set.seed(1234)

n <- 10000

accepted <- numeric(0) # initialise numeric variables/vectors

while(length(accepted) < n){ # keep sampling until we have n accepted values
  x <- runif(1) # proposal: X ~ Uniform(0,1)
  y <- runif(1, 0, 3) # auxiliary variable: Y ~ Uniform(0, 3)

  if(y <= 3 * x^2){ # accept if point lies under the curve f(x)
    accepted <- c(accepted, x) # keep accepted x
  } # otherwise reject and try again
}

hist(accepted, prob = TRUE, col = "skyblue", main = "Acceptance-Rejection Sampling")
lines(density(accepted), col = "blue", lwd = 2) # Kernel Density Estimate (KDE)
curve(3 * x^2, col = "red", lwd = 2, add = TRUE) # True Density

legend("topright", legend = c("Histogram", "KDE", "True density"),
       col = c("skyblue", "blue", "red"),
       lwd = c(NA, 2, 2), pch = c(15, NA, NA), bty = "n")
```



- The histogram matches the increasing shape of the target density, so larger values near 1 are more likely.
- Small differences between bars and the curve are due to randomness and binning.

From Rectangles to General Envelopes

In the first example, we sampled uniformly from a rectangle and accepted points under the curve.

That works well when:

- the support is simple
- the density is bounded
- a rectangle is easy to use

But many target densities are more complicated. Therefore, we need a more flexible idea.

Key Idea

Suppose we want to simulate from a target density $f(x)$. Instead of sampling from a rectangle, we choose:

- an easier density $g(x)$
- a constant $M > 0$

such that

$$\boxed{f(x) \leq Mg(x) \quad \forall x}$$

Then $Mg(x)$ acts as an **envelope** over $f(x)$.

👉 The closer $Mg(x)$ is to $f(x)$, the more efficient the method.

General Acceptance-Rejection Algorithm

To simulate $X \sim f(x)$:

Generate a candidate $Y \sim g(x)$ → **Generate** $U \sim \text{Uniform}(0,1)$ → **Accept** Y **if** $U \leq \dfrac{f(Y)}{Mg(Y)}$ → **Otherwise reject** Y **and repeat**

- If the candidate Y is accepted, then we set $X = Y$.
- The accepted values follow the target density $f(x)$.

Efficiency of the Method

The probability of acceptance is

$$\frac{1}{M}.$$

- smaller M $\rightarrow$ higher acceptance rate
- larger M $\rightarrow$ more rejections

Choosing a Good Proposal Density

- have the same support as $f(x)$
- be easy to sample from
- make M as small as possible

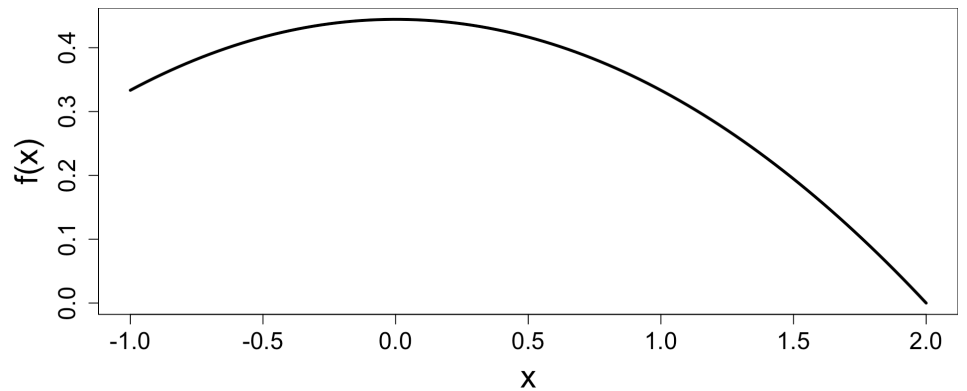
- Uniform proposal on a bounded interval
- Exponential proposal for right-skewed targets
- Normal proposal for bell-shaped targets

Cubic Polynomial CDF Example (From Part 1)

The measurement error X (in mV) of a particular volt-meter has the following distribution:

$$f(x) = \begin{cases} \dfrac{4 - x^2}{9}, & -1 \le x \le 2 \\ 0, & \text{otherwise.} \end{cases}$$

Simulate $f(x)$ using acceptance-rejection method.



Step 0: Check whether $f(x)$ is valid

$$\int_{-1}^{2} \frac{4 - x^2}{9}\, dx = 1 \rightarrow \text{ valid PDF.}$$

Step 1: Determine $Mg(x); f(x) \le Mg(x)$

Because the support is bounded, the best first proposal is:

$$g(x) = \text{Uniform}(-1, 2) = \frac{1}{3}, \quad -1 \le x \le 2.$$

Since $g(x) = 1/3$, this becomes

$$\frac{4 - x^2}{3} \le M$$

Now on [-1, 2], the maximum of LHS occurs at $x = 0$,

$$\max_{-1 \le x \le 2} \frac{4 - x^2}{3} = \frac{4}{3}$$

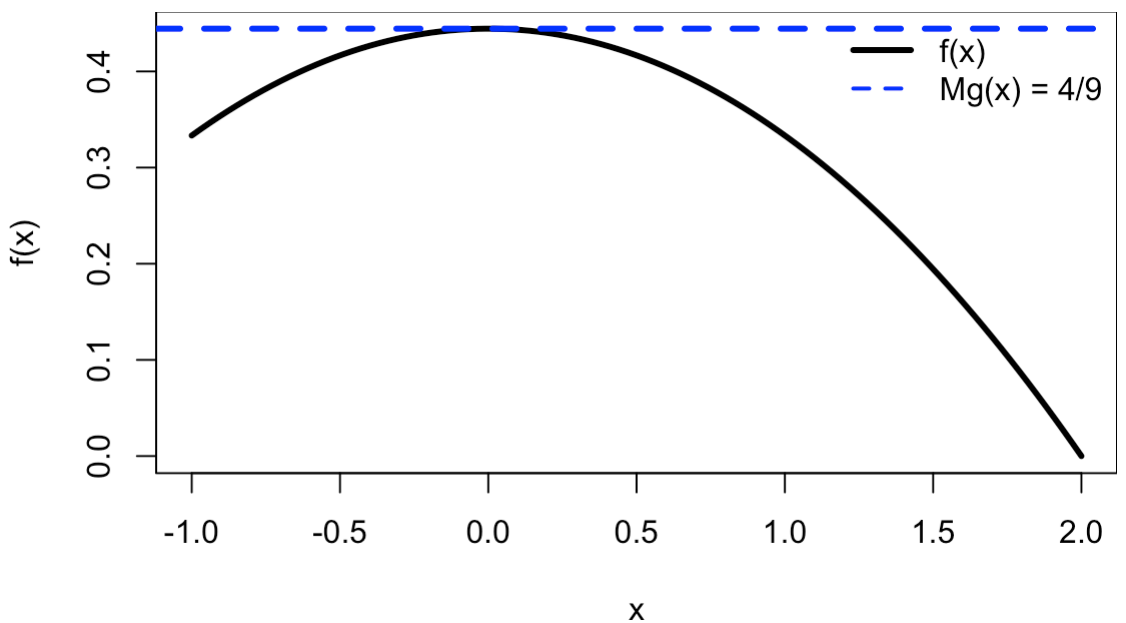
Hence, we can choose

$$M = \frac{4}{3} \quad \text{with an acceptance rate of} \quad \frac{1}{M} = \frac{3}{4} = 75\%$$

This gives

$$Mg(x) = \frac{4}{3} \cdot \frac{1}{3} = \frac{4}{9}$$

- So the envelope is just a horizontal line at height 4/9.



Step 2: Determine the acceptance rule

The general rule is

$$U \leq \frac{f(Y)}{Mg(Y)}$$

where $Y \sim g$ and $U \sim \text{Uniform}(0, 1)$.

Substitute in, gives

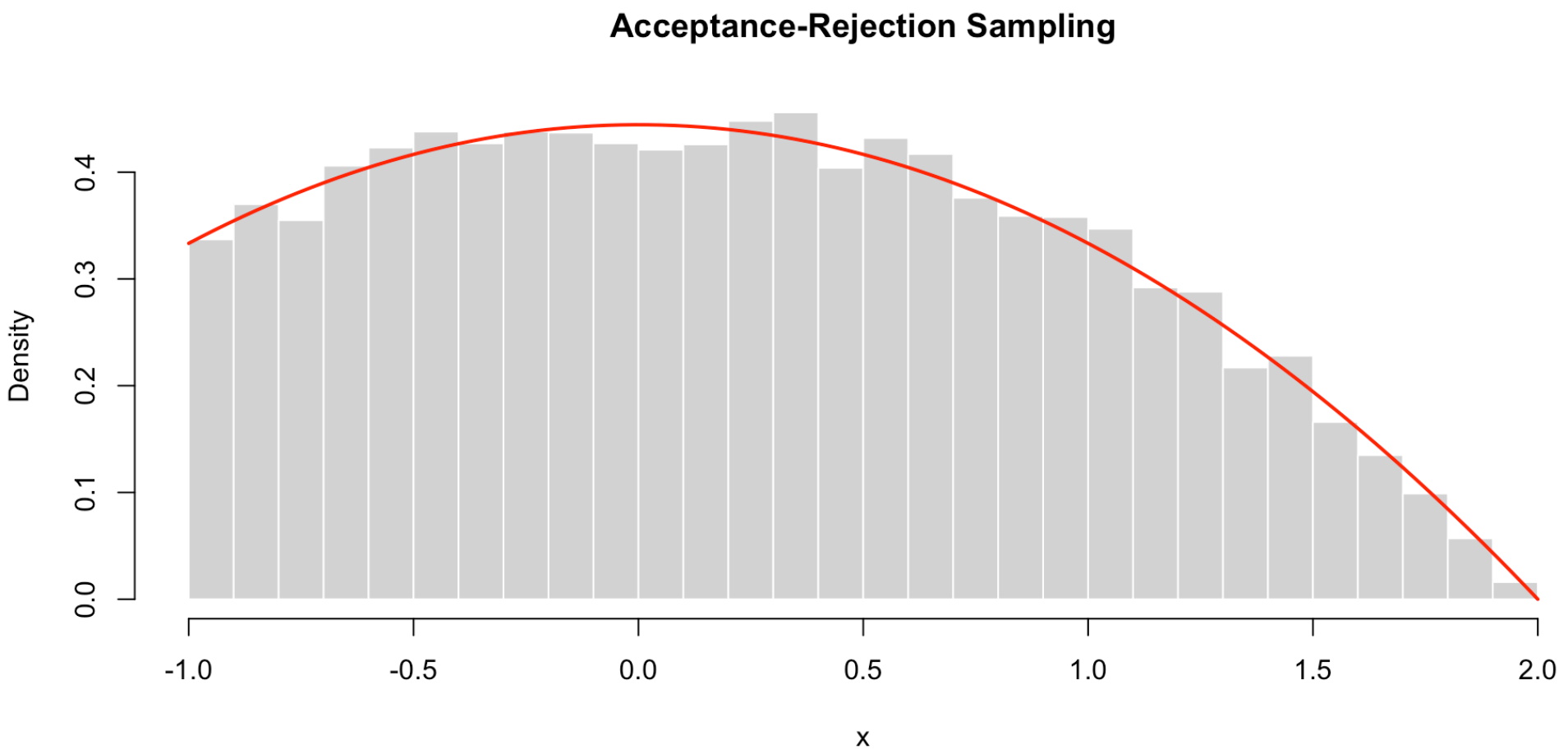
$$\frac{f(Y)}{Mg(Y)} = \frac{4 - y^2}{4}$$

So accept Y if

$$U \leq \frac{4 - y^2}{4}.$$

```r
set.seed(1234)
n <- 10000
x <- numeric(0) # store accepted samples

while(length(x) < n){ # keep sampling until we have n accepted values
  y <- runif(n, min = -1, max = 2)   # proposal sample Y ~ g(x) ~ Uniform(-1, 2)
  u <- runif(n)                      # U ~ Uniform(0,1) for acceptance test
  accept <- u <= (4 - y^2)/4         # acceptance rule (returns TRUE/FALSE)
  x <- c(x, y[accept])               # keep accepted y
}

x <- x[1:n] # trim to exactly n samples
```

```r
hist(x, prob = TRUE, breaks = 40,
     col = "lightgray", border = "white",
     main = "Acceptance-Rejection Sampling",
     xlab = "x")

curve((4 - x^2)/9, from = -1, to = 2,
      add = TRUE, col = "red", lwd = 2)
```

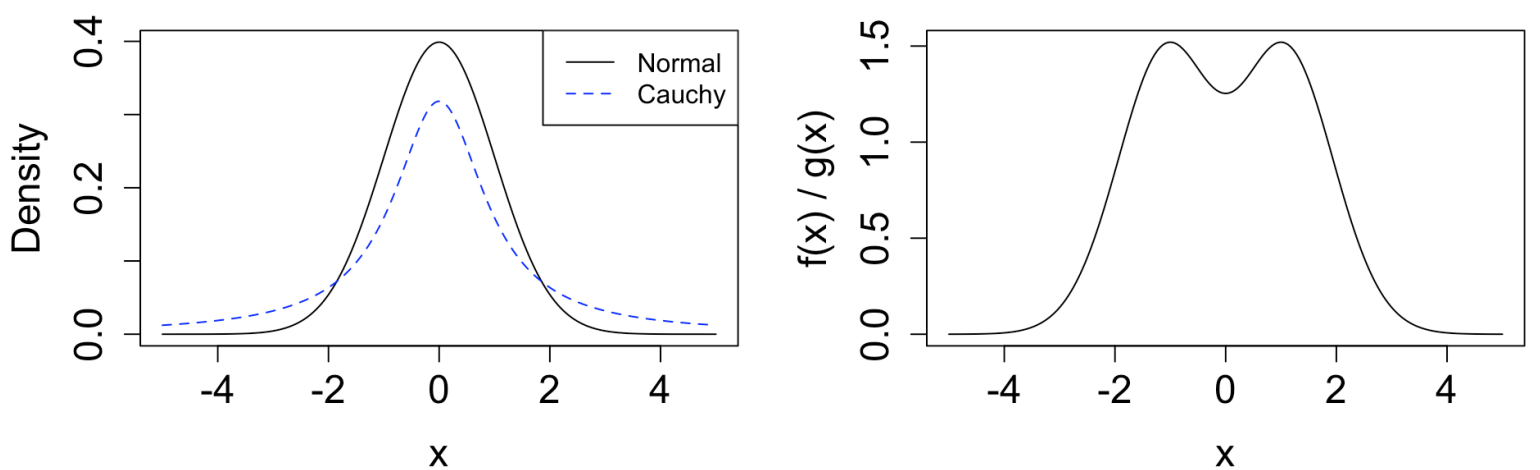


Normal-Cauchy Example

> Generate the standard normal distribution using the standard Cauchy distribution
>
> Recall the PDF of $X \sim N(0,1)$:
>
> $$f(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right), \quad x \in \mathbb{R}.$$
>
> The PDF of $X \sim \text{Cauchy}$:
>
> $$g(x) = \frac{1}{\pi(1+x^2)}, \quad x \in \mathbb{R}.$$

Let's compare the distributions:

▶ Code



- From RHS, the standard cauchy distribution is wider but shorter than the standard normal distribution.
- From LHS, the $f(x)/g(x)$ ratio is maximised around 1.5.

Step 1: Find M

$$\frac{f(x)}{g(x)} = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right) \cdot \pi(1+x^2) = \frac{\pi}{\sqrt{2\pi}}(1+x^2)\exp\left(-\frac{x^2}{2}\right)$$

$$\max_{-\infty<x<\infty} (1+x^2)\exp\left(-\frac{x^2}{2}\right)$$

Differentiate and set it equal to 0:

$$\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = 2x\exp\left(-\frac{x^2}{2}\right) + (1+x^2)(-x)\exp\left(-\frac{x^2}{2}\right)$$

$$0 = \exp\left(-\frac{x^2}{2}\right)x(1-x^2)$$

yields three critical points: $x = 0, x = \pm 1.$

- From the ratio plot, we would expect the maximum to occur at $x = \pm 1.$
- If this is not obvious, use the second derivative test to verify.

Substitute $x = \pm 1$ into $f(x)/g(x)$ yields

$$M = 2e^{-1/2}\sqrt{\frac{\pi}{2}} \approx 1.5203 \rightarrow \text{ with the acceptance probability of } 0.6577.$$

Step 2: Determine the acceptance rule

The general rule is

$$U \leq \frac{f(Y)}{Mg(Y)}$$

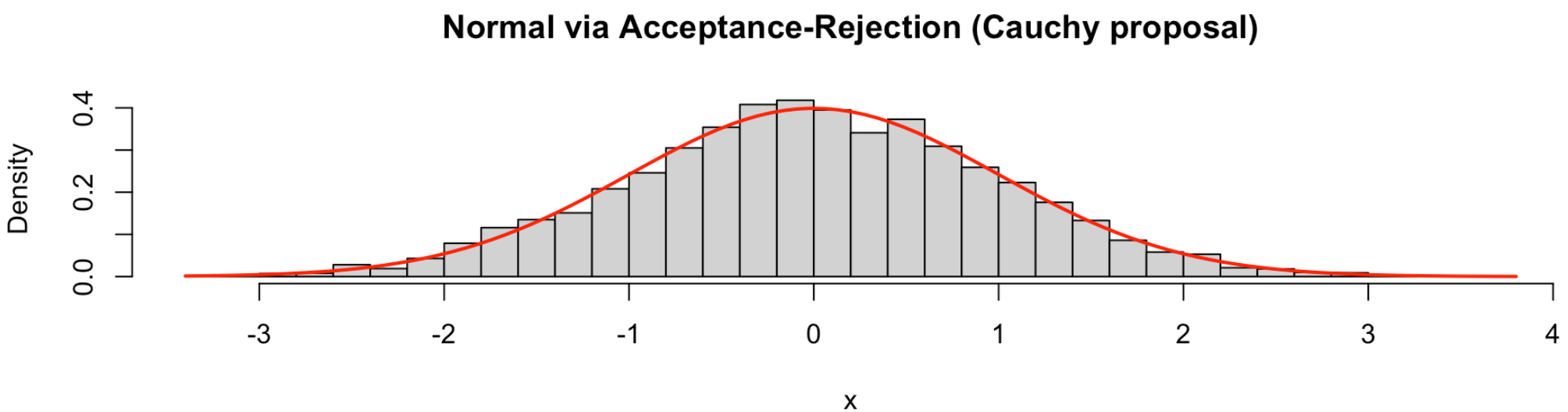
where $Y \sim g \sim \text{Cauchy}$ and $U \sim \text{Uniform}(0, 1)$.

Substitute in, gives the acceptance rule

$$U \leq \frac{f(Y)}{Mg(Y)} = \frac{\sqrt{e}}{2}(1 + y^2)\exp\left(-\frac{y^2}{2}\right)$$

```r
set.seed(1234)
n <- 5000                # number of samples needed
M <- 2 * exp(-1/2) * sqrt(pi/2)   # exact M ≈ 1.5203
samples <- numeric(0)

while(length(samples) < n){
  y <- rcauchy(n) # propose from Cauchy
  u <- runif(n)   # uniform for acceptance
  f <- dnorm(y)   # target density
  g <- dcauchy(y) # proposal density

  accept <- u <= f / (M * g) # acceptance rule
  samples <- c(samples, y[accept])  # keep accepted values
}

samples <- samples[1:n] # trim to exactly n samples
```

```r
hist(samples, prob = TRUE, breaks = 40,
     col = "lightgray",
     main = "Normal via Acceptance-Rejection (Cauchy proposal)",
     xlab = "x")

curve(dnorm(x), add = TRUE, col = "red", lwd = 2)
```




```r
set.seed(1234)
y <- rcauchy(5000); u <- runif(5000)
M <- 2 * exp(-1/2) * sqrt(pi/2)
accept <- u <= dnorm(y) / (M * dcauchy(y))
cat("The empirical acceptance rate is", mean(accept))
```

```
The empirical acceptance rate is 0.6578
```

Part 3: Precision of Simulation Result

Up until this point, we learned how to generate random variables using

- known distributions
- inverse transform
- acceptance-rejection

We can now simulate almost any distribution.

How accurate are our simulation results?

- If we simulate 1,000 samples… is that enough?
- What about 10,000?
- How close are we to the true value?

When we use these simulations to estimate quantities,

this is called **Monte Carlo estimation**.

Monte Carlo Estimation

Suppose we want to estimate:

$$\mu = \mathbb{E}[X]$$

We simulate i.i.d random variables:

$$X_1, X_2, \ldots, X_n$$

and compute:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^{n} X_i,$$

where n is the number of simulated samples.

Key Result (CLT)

The CLT tells us about the distribution of $\bar{X}_n$:

$$\bar{X}_n \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$$

based on the same n samples.

> The CLT tells us that the error of Monte Carlo estimate is approximately Normally distributed for large n , which allows us to construct **confidence intervals** and *quantify simulation precision.*

Monte Carlo Standard Error

The **standard error** measures simulation uncertainty:

$$\mathrm{SE}(\bar{X}_n) = \frac{\sigma}{\sqrt{n}}$$

In practice:

$$\widehat{\mathrm{SE}} = \frac{s}{\sqrt{n}}$$

where s is the sample standard deviation.

Interpretation: Larger n $\rightarrow$ smaller error, in which error decreases at rate $1/\sqrt{n}$.

```r
set.seed(1234)

n <- 1000
x <- rexp(n, rate = 2)
x2 <- rexp(4*n, rate = 2)

mean(x); mean(x2)
```

```
[1] 0.5003067
[1] 0.5012854
```

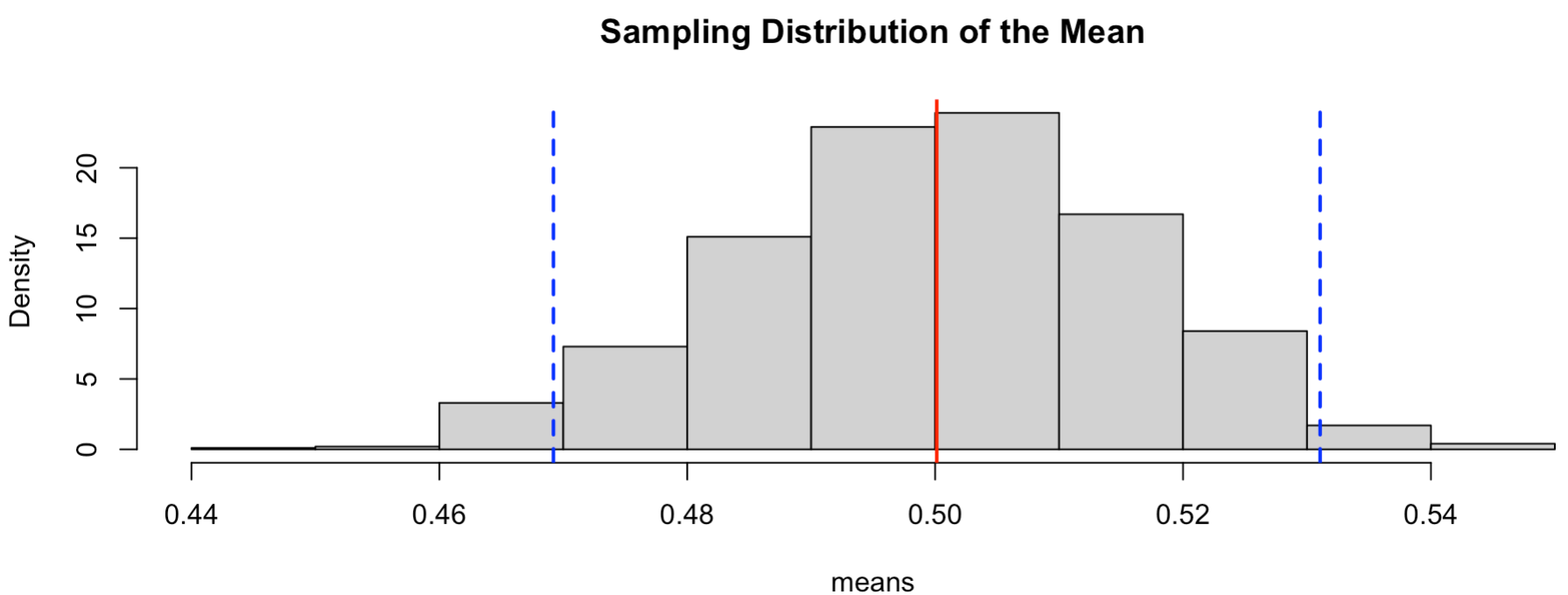
```r
sd(x)/sqrt(n); sd(x2)/sqrt(4*n)
```

```
[1] 0.01599075
[1] 0.007821783
```

Confidence Interval for the Mean

CI for sample mean is

$$\bar{X}_n \pm z_{\alpha/2}\frac{s}{\sqrt{n}}$$

Confidence Level $(1-\alpha)100\%$	$z_{\alpha/2}$
90%	1.645
95%	1.960
99%	1.645

▶ Code